

Blister, bubbles and protrusions in 2D materials

A working note on toy molecular dynamics, AIREBO and LAMMPS —
what each model is, what it can be trusted for, and how to check it

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Abstract

This note covers three things that turned out to be one thing. First, what the DSW `graphene-md` toy model actually is — Morse bonds, angular springs and a Lennard-Jones substrate — and how that differs from AIREBO and from what LAMMPS brings. Second, the mechanics of pressurised blisters in 2D membranes, from the Hencky solution through the adhesion balance that Bunch and co-workers used to measure the adhesion energy of graphene. Third, and most usefully, a method for validating a fast approximate model against an expensive accurate one, including the ways such a validation can look convincing while proving nothing. All measurements quoted are from runs performed for this note.

Contents

1	The two models	2
1.1	The toy model: Morse plus angular springs plus Lennard-Jones	2
1.2	AIREBO: a bond-order potential	2
1.3	What each is good for	3
2	What LAMMPS adds over a toy MD	3
3	Blister mechanics	4
3.1	The Hencky solution	4
3.2	The adhesion balance	4
3.3	Constant pressure versus constant N	4
3.4	How the pressure is actually implemented, and how far to trust it	5
4	What we measured	6
4.1	The exporter reproduces the plugin	6
4.2	Blister geometry across engines	6
4.3	A stable blister, and Γ	7
4.4	The rule tested at a different size	7
5	Twisted bubbles: registry and strain	8
5.1	A single sheet, with a free peel front	8
5.2	A genuine bilayer, and where the gas sits	8
6	How to validate a toy model against LAMMPS	9
6.1	The tests, strongest first	9
6.2	Five ways a validation lies	10
6.3	A working procedure	11

7	Practical notes	11
A	A worked LAMMPS deck, line by line	11

1 The two models

1.1 The toy model: Morse plus angular springs plus Lennard-Jones

Three ingredients, each a pair or triplet interaction with an explicit formula.

Morse bonds. Nearest neighbours interact through

$$U_M(d) = D_e \left[\left(1 - e^{-\alpha(d-r_e)} \right)^2 - 1 \right], \quad (1)$$

a spring near r_e that softens and goes flat at large separation. Because it flattens, it can represent a bond *breaking*: past a chosen extension the restoring force falls away. The neighbour list is fixed at build time, so the model knows which pairs are “bonded” and never discovers new ones except through an explicit re-forming rule.

Angular springs. A honeycomb held together only by central forces between nearest neighbours has *no shear rigidity* — it is a mechanism, not a solid, and it collapses. The toy model restores stiffness with harmonic springs to second neighbours at $\sqrt{3}r_e$ (equivalently, a bond-angle term). This is a device, not chemistry: it produces a stable lattice with roughly the right elastic constants and nothing more.

Lennard-Jones substrate. The sheet is attracted to a rigid substrate by a 12–6 potential, truncated at 3σ . This provides adhesion, and it is the term that decides whether a blister stays stuck down.

Bending. A crude “umbrella” term penalising out-of-plane displacement of an atom relative to its neighbours. It has no LAMMPS analogue and is the one term the plugin’s LAMMPS exporter cannot translate.

1.2 AIREBO: a bond-order potential

AIREBO is a different kind of object. Its core is REBO, a *bond-order* potential in the Tersoff–Brenner family:

$$U = \sum_i \sum_{j>i} \left[V^R(r_{ij}) - b_{ij} V^A(r_{ij}) \right], \quad (2)$$

a repulsive term minus an attractive term whose strength b_{ij} depends on the *local environment* — how many neighbours atom i has, at what angles, and how their own bonding looks. That single feature is the difference that matters:

- **There is no bond list.** Bonding is recomputed from geometry every step. Atoms form and break bonds because the environment changed, not because a distance crossed a threshold.
- **Coordination is energetic.** A three-fold carbon and a two-fold carbon are genuinely different, so edges, vacancies and reconstructions come out of the potential rather than being imposed.
- **Angles are implicit.** No separate angular spring is needed; the bond order carries the angular dependence.

AIREBO adds to REBO an *adaptive* Lennard-Jones term (switched off between bonded neighbours, on between non-bonded ones, so interlayer vdW works without corrupting covalent bonding) and a torsional term for dihedral angles. Hence **Adaptive Intermolecular REBO**.

The price is cost: the bond order requires neighbour-of-neighbour sums, so AIREBO is one to two orders of magnitude more expensive per atom-step than a pair potential with a fixed bond list.

1.3 What each is good for

Question	Toy model	AIREBO
membrane stretching, blister shape	reliable	reliable
adhesion-driven draping, wrinkling	reliable	reliable
phonons, elastic constants	reliable if fitted	reliable
bending near a peel front	crude	reliable
bond breaking, tearing, edges	a distance rule, not chemistry	reliable
defects, reconstruction, chemistry	no	reliable
speed	25 steps/s at 233k atoms	9 steps/s

Table 1: The split is not “approximate vs exact” but “mechanics vs chemistry”. Where the physics is membrane mechanics, the cheap model is not an approximation to the expensive one — it contains the same physics.

2 What LAMMPS adds over a toy MD

A purpose-built simulation does one thing quickly. LAMMPS is a framework, and the difference shows up in five places.

1. **A potentials library.** AIREBO, ExTeP, REBO-MoS₂, Stillinger–Weber, ReaxFF, machine-learned potentials — all published, parameterised and tested. Writing one yourself is a research project.
2. **Fixes as composable physics.** Thermostats and barostats, real ensembles (NVE/NVT/NPT), constraints, external forces, moving boundaries, bond breaking and creation. Each is a well-tested module you switch on.
3. **Neighbour machinery.** Binning, skins, rebuild criteria, exclusion lists, and the diagnostics to see when it goes wrong.
4. **Scale.** MPI domain decomposition to thousands of cores; the toy model is one shared-memory node.
5. **Analysis.** Computes, ensemble averages, stress per atom, correlation functions, dumps in formats other tools read.

What it costs is generality overhead, and it is measurable. On the identical Morse system LAMMPS spent **43 % of its time in Modify — fix move variable** evaluating an `exp()` per substrate atom per step — against 38 % in **Pair**. On the identical AIREBO system, **Pair rose to 73 %** and **Modify** fell to 23 %. The framework overhead is a fixed cost that stops mattering once the potential is genuinely expensive. Correspondingly the plugin’s speed advantage falls from $1.9\times$ on the toy model to $1.15\times$ on AIREBO.

3 Blister mechanics

3.1 The Hencky solution

A clamped circular membrane of radius a under uniform pressure p , with 2D modulus E_{2D} , deflects at its centre by δ where

$$\frac{p a}{E_{2D}} = K(\nu) \left(\frac{\delta}{a} \right)^3, \quad K \approx 3.09 \quad (\nu = 0.16), \quad (3)$$

with shape $w(\rho) = \delta (1 - \rho^2)^{2/3}$, $\rho = r/a$, and enclosed volume

$$V = C \pi a^2 \delta, \quad C \approx 0.52. \quad (4)$$

The cubic in δ is the signature of a *membrane*: stiffness comes from stretching, not bending, so the structure gets stiffer as it deflects.

3.2 The adhesion balance

Blisters are not clamped — they are stuck down, and they can peel. Integrating $p dV$ at fixed a using (3) gives the stored strain energy as exactly one quarter of the pressure–volume work, so the potential energy is $\Pi = -\frac{3}{4}pV$. The energy release rate at the delamination front follows from $G = -(2\pi a)^{-1} \partial \Pi / \partial a$:

$$G = \frac{5}{4} C p \delta \approx 0.65 p \delta \quad (5)$$

Setting $G = \Gamma$, the adhesion energy per unit area, and eliminating p with (3):

$$\frac{\delta}{a} = \left[\frac{4\Gamma}{5 C K E_{2D}} \right]^{1/4} \approx \left(\frac{\Gamma}{680 \text{ N/m}} \right)^{1/4} \quad (6)$$

and, at that aspect ratio,

$$p \propto \frac{\Gamma^{3/4} E_{2D}^{1/4}}{a} \quad (7)$$

Two consequences worth internalising. The aspect ratio is **size-independent** — no length appears in (6) — which is why measured graphene bubbles cluster around $\delta/a \sim 0.1$ – 0.2 across orders of magnitude in radius. And it depends on adhesion only as a **fourth root**, so a blister is a blunt instrument for measuring Γ : a factor of two error in δ/a is a factor of sixteen in Γ .

This is the basis of the constant- N blister test of Koenig, Boddeti, Dunn and Bunch, who used it to obtain $\Gamma = 0.45 \text{ J/m}^2$ for monolayer graphene on SiO_2 and 0.31 J/m^2 for two-to-five layers.

3.3 Constant pressure versus constant N

If the pressure is held fixed, inflation is unstable: the gas does unlimited work. If instead a *fixed number of molecules* is trapped, then $p = Nk_B T/V$, and inflating raises V and drops p . That negative feedback is what makes a blister an equilibrium object rather than a runaway.

This matters for simulation, and it was the origin of a bug in ours: the protrusion driver applied a *prescribed* pressure schedule, so the blister inflated without limit and was still rising when the schedule had returned the pressure to zero. Replacing it with a fixed- N gas law, measuring V from the sheet each step, made the pressure fall from 2492 to 557 MPa as the blister grew — the feedback appearing where before there was none.

3.4 How the pressure is actually implemented, and how far to trust it

The gas is not simulated. It is a boundary condition, applied as follows: at each step the pocket volume is measured from the sheet,

$$V = \sum_{i:r_i < R} \max(z_i - z_0, 0) A_{\text{atom}}, \quad (8)$$

the pressure is taken from the isothermal ideal gas law $p = Nk_B T/V$ at fixed $T = 300$ K, and a *uniform vertical* force pA_{atom} is added to every sheet atom inside a hard-edged disc of radius R . N is fixed once, from the requested pressure at the Hencky equilibrium volume.

Six approximations are buried in that paragraph, in decreasing order of how much they should worry you.

1. The ideal gas law is unphysical at these pressures. At 1 GPa and 300 K, $n = p/k_B T = 241 \text{ nm}^{-3}$ — roughly *fourteen times* the number density of liquid nitrogen. No real gas can be that dense; the molecules do not fit. So the “ N molecules” in the model are a bookkeeping device, not a physical count.

What survives is the *stiffness law*: the model asserts $p \propto V^{-1}$, i.e. $dp/dV = -p/V$. A real fluid at liquid-like density is far stiffer than that — excluded volume makes p rise much faster as V shrinks. Our blister is therefore **too soft against volume change**. Practically: a real gas would resist the initial inflation more, snap through less violently, and ring down differently. A van der Waals form, $p = Nk_B T/(V - Nb) - a(N/V)^2$, would cost one line and remove most of the error.

2. The footprint is fixed and cannot follow a peel front — and peeling is the whole mechanism. The gas pushes only inside $r < R$, always. This is not one limitation among several: **the blister test measures Γ because the contact line moves**. Equation (5) comes from $\partial\Pi/\partial a$ at a free delamination front, so the radius must be an *output* of the calculation. Pinning it is differentiating with respect to a variable the model does not allow to vary.

Two things save the result in practice, and they are worth separating.

First, **the peel is present, just not where the gas is**. The sheet outside R is genuinely free: lifted by membrane tension, held down by adhesion, with a contact line that finds its own radius. That is why the blister behaves as though its radius were 22 % larger than the footprint. What is missing is not the moving contact line but the gas loading over the newly opened annulus.

Second, **we can check the answer, which an experiment cannot**. Koenig *et al.* had to trust the blister analysis because Γ was the unknown quantity. Here Γ is an *input*: integrating the 12–6 substrate potential over a continuum sheet gives

$$\Gamma_{\text{LJ}} = 1.2 \pi n_t n_s \varepsilon \sigma^2 = 0.246 \text{ J/m}^2 \quad (0.241 \text{ with the } 3\sigma \text{ cutoff}), \quad (9)$$

at an equilibrium separation of exactly $d = \sigma = 3.42 \text{ \AA}$. The blister test returned 0.22 J/m^2 — **9 % low**, with the sign one would predict: a pinned footprint pressurises less area than a true blister of the same delaminated radius, so $p\delta$ slightly under-counts G .

That the cutoff costs only 2.1 % of the adhesion energy also disposes of a plausible-sounding worry: extending the Lennard-Jones range would not meaningfully change any threshold in this note.

The fix. Replace the $r < R$ test with a *gap* test: pressurise wherever the sheet stands more than about a molecular diameter above contact, and count that volume. The gas then occupies the void it actually occupies, the pressurised region follows the delamination front automatically, and the blister radius becomes an output. A small pre-lifted patch is needed to nucleate, since a perfectly adhered sheet encloses no volume. With that change the model would be a blister test by construction rather than by accident, and the 9 % could be re-measured to see whether it closes.

3. Volume is counted only inside the footprint. Any lift outside R contributes gas volume in reality but not in the model, so p is correspondingly overestimated. Consistent with (2), and in the same direction.

4. The force is vertical, not normal to the surface. Real pressure acts along the local surface normal. At $\delta/a \approx 0.13$ the steepest part of the profile is inclined by roughly 20–25°, so near the rim the model over-counts the vertical component by $\sim 10\%$ and omits an inward radial component of comparable size. The rim is exactly where the adhesion balance is decided, so this is not as harmless as its magnitude suggests.

5. Isothermal, with no thermal coupling. T is pinned at 300 K. A real snap-through expansion is closer to adiabatic: the gas would cool, p would fall further than V^{-1} alone predicts, and the overshoot would be smaller. Our ring-down is more violent than reality.

6. No leakage — and here the model is right. Fixed N assumes a perfect seal. For graphene that is a genuinely good idealisation: Bunch *et al.* showed graphene membranes to be impermeable to standard gases, which is what makes the constant- N blister test possible experimentally in the first place.

What this does *not* contaminate. The adhesion energy extracted in Section 4.3 is safe from all of the above, because $G = \frac{5}{4}Cp\delta$ uses the pressure and deflection *measured at equilibrium*. Whatever route the gas took to get there, and whatever equation of state generated that pressure, the balance at equilibrium involves only p and δ . The gas law determines *which* equilibrium you land on; it does not change the condition that defines it. Results that depend on the path — the snap-through amplitude, the ring-down, the dynamic threshold pressure — are a different matter and should be treated as model behaviour.

4 What we measured

4.1 The exporter reproduces the plugin

The strongest available check on two implementations is to put them in the *same state* and compare an extensive quantity. Exporting a LAMMPS deck from the plugin and evaluating step 0:

System	Plugin E_{pot}	Stock LAMMPS	rel. difference
930 629 atoms, Morse	−14591.6 eV	−14591.569 eV	2×10^{-6}
232 735 atoms, Morse	−3655.11 eV	−3655.106 eV	1×10^{-6}
232 735 atoms, AIREBO	−711418 eV	−711418.41 eV	6×10^{-7}

AIREBO’s cohesive energy comes out at -7.44 eV/atom against the published -7.408 for infinite graphene, the balance being substrate adhesion — an independent check that the potential is correctly parameterised.

4.2 Blister geometry across engines

For the twisted-bubble case the four engines split cleanly:

- **In-plane** quantities are engine-insensitive. Lattice registry statistics agreed to four decimal places (mean 0.6668, s.d. 0.2717) between the plugin’s embedded AIREBO and stock LAMMPS, and the atoms differed by at most 0.097 Å in plane.

- **Out-of-plane** quantities are not. The same two runs differed by $1.58 \text{ \AA}$ vertically, and blister heights differed by up to 15 %.

4.3 A stable blister, and Γ

With the fixed- N gas, a stable blister exists in a narrow window. At $R = 3 \text{ nm}$ and 700 MPa the volume drift fell to 1.7 % over 49 000 steps, settling at $\delta = 5.13 \text{ \AA}$, $\delta/a = 0.171$, at a settled pressure of 658 MPa . Applying (5):

$$\Gamma = 0.65 p \delta = 0.65 \times 658 \times 10^6 \times 5.13 \times 10^{-10} = 0.22 \text{ J/m}^2, \quad (10)$$

The number to compare against is not Koenig’s experimental value but the adhesion energy we *put into* the model, $\Gamma_{\text{LJ}} = 0.241 \text{ J/m}^2$ (Section 3.4): the blister test recovers its own input to **9 %**, and low, as a pinned footprint should make it. It also happens to sit near the 0.24 J/m^2 Koenig *et al.* report, but that is a different interface and the closeness is a coincidence of parameterisation, not evidence.

Feeding that Γ back through (6) predicts runaway near 250 MPa at $R = 10 \text{ nm}$; the measured transition was $200\text{--}260 \text{ MPa}$.

4.4 The rule tested at a different size

The strongest test of (6) is prediction rather than fitting. Taking $\Gamma = 0.22 \text{ J/m}^2$ from the $R = 3 \text{ nm}$ blister above, the rule predicts $\delta/a = 0.134$ at *any* radius. Running $R = 10 \text{ nm}$ at 160 MPa — a different size, and a quarter of the pressure — gives a settled $\delta/a = 0.128$ (Figure 1), within 4 % of the prediction.

The approach to equilibrium is itself instructive. The sheet stays pinned flat while the gas fills, then delaminates in a *snap-through* at around step 10 000, overshoots to $24 \text{ \AA}$, and rings down to $12.84 \pm 1.8 \text{ \AA}$ over the following 60 000 steps, the oscillation amplitude decaying by a factor of fourteen. The snap is physical: adhesion pins the membrane until the gas load exceeds the peel threshold, and then releases abruptly.

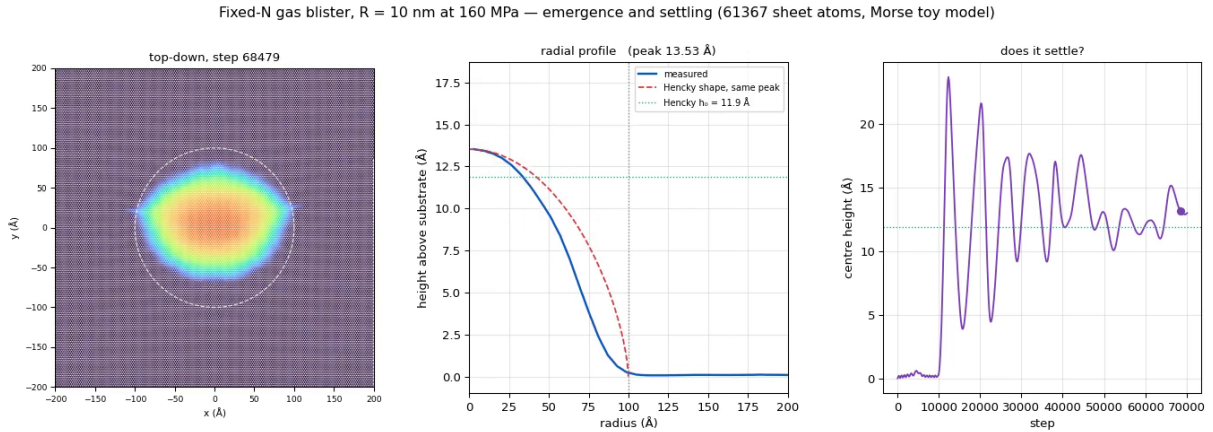


Figure 1: A settled fixed- N blister, $R = 10 \text{ nm}$ at 160 MPa . Left: plan view, dashed circle marking the gas footprint. Centre: the measured radial profile against the Hencky shape through the same peak — it tracks near the centre and falls below from $r \approx 40 \text{ \AA}$ outward, as a peeling rim should. Right: centre height against step, showing the snap-through and the ring-down.

A measurement artefact worth naming. The automated settling test reported “still moving” with 13 % drift, because it compared the means of two successive windows — a statistic that a *decaying oscillation* defeats. The mean was in fact stationary (a linear fit over the last half gives

$-0.55 \text{ \AA}$ across 35 000 steps) while the amplitude fell from 6.05 to 0.43 $\text{\AA}$. The system had settled; the test could not see it. This belongs with the failure modes of Section 5.2.

The catch. Extracting Γ the *other* way — from the measured aspect ratio through (6) — gives 0.58 J/m^2 , a factor 2.6 different. The two routes disagree because our blister does not satisfy (3): it sits 31 % above the Hencky height at $R = 3 \text{ nm}$. The Koenig extraction assumes the membrane obeys Hencky. When it does not, the choice of extraction route silently changes the answer threefold, and nothing in either calculation warns you.

The discrepancy is a *boundary condition*, not a modulus. Hencky clamps the rim; a real blister peels, so the effective radius exceeds the gas footprint, and since $\delta \propto a^{4/3}$ a 22 % larger effective radius produces exactly the observed 31 % excess. Two pieces of evidence support this over the obvious alternative that the toy model is too soft. First, running the identical case on AIREBO — which has the correct graphene stiffness — gave 5.105 $\text{\AA}$ against the toy model’s 5.131 $\text{\AA}$, agreeing to 0.5 %. Second, the excess *shrinks with size*: +31 % at $R = 3 \text{ nm}$ but +8.3 % at $R = 10 \text{ nm}$, exactly as a rim correction should.

5 Twisted bubbles: registry and strain

5.1 A single sheet, with a free peel front

Replacing the fixed footprint with one that follows the peel front (Section 3.4) makes the delamination radius an output. For a 2° twisted sheet with a 4 nm seed at 600 MPa the front settled at 45 $\text{\AA}$ against an adhesion-limited prediction of 42.6 $\text{\AA}$.

The distortion is not where one might guess. The bubble interior is in mild tension ($\text{tr } \varepsilon \approx +0.017$ at the centre), but the dominant features are a **ring of compression reaching -0.045 and a ring of shear reaching 0.031 sitting exactly at the delamination edge**, with the adhered region beyond essentially unstrained. That is the classic blister signature — radial tension, azimuthal compression at the rim — resolved atom by atom.

The moiré is also visibly smeared inside the bubble. Registry depends only on in-plane position, so that is real lateral displacement: the sheet stretches radially to accommodate the bulge and slides out of registry with the substrate.

5.2 A genuine bilayer, and where the gas sits

With two live layers the placement of the gas changes the answer completely. Identical twist, seed and pressure; only **gasWhere** differs.

Placement	peak height ($\text{\AA}$)		dilatation range	
	layer 1	layer 2	layer 1	layer 2
gas between the layers	3.53	12.35	$-0.003 \cdots + 0.001$	$-0.023 \cdots + 0.006$
gas below the bilayer	7.54	10.94	$-0.011 \cdots + 0.003$	$-0.011 \cdots + 0.003$

Table 2: Gas between the layers puts essentially all the deformation into the upper layer — the lower one stays flat and ten times less strained. Gas below deforms both almost identically.

Two consequences follow. First, the below case is **1.4 $\text{\AA}$ lower at the same pressure** (10.94 against 12.35 $\text{\AA}$), because two layers are being bent rather than one: the effective bending rigidity is higher. Second, the interlayer gap behaves quite differently:

A threshold that differs from the monolayer. At 600 MPa the contact line stayed pinned at the seed in both bilayer runs, where a single sheet at the same seed and pressure advanced from 40 to 45 $\text{\AA}$. Raising the pressure to 1400 MPa moved the bilayer front out to 62 $\text{\AA}$. The plausible

	gap at the bubble centre	gap in the far field
gas between, 600 MPa	8.28 Å	3.42 Å
gas between, 1400 MPa	13.04 Å	3.42 Å
gas below, 600 MPa	3.40 Å	3.42 Å
gas below, 1400 MPa	3.41 Å	3.42 Å

Table 3: A bubble between the layers prisms them apart, so $h \gg 3.35$ Å over the bubble is physics, not an artefact. A bubble below carries the pair up together and the spacing never departs from equilibrium, even with the bilayer lifted 10.7 Å off the substrate. The far field is 3.42 Å in every run, which is the built-in 3.35 Å plus thermal roughness.

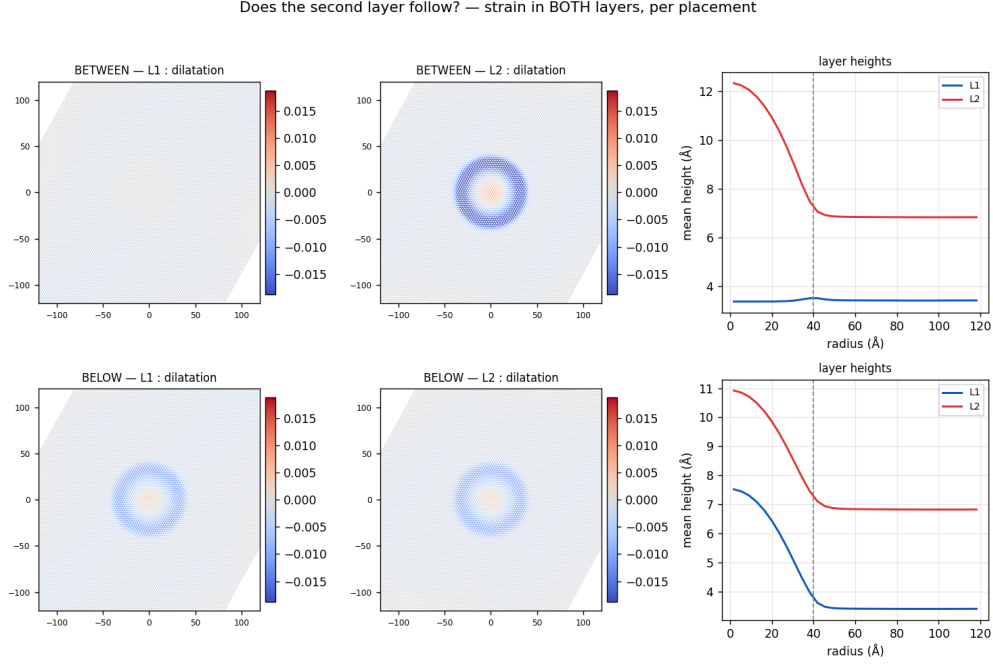


Figure 2: Dilatation in both layers for each placement, with the layer height profiles. Top row, gas between: the strain is confined to layer 2. Bottom row, gas below: the two layers carry nearly identical strain and their height profiles are parallel curves a constant 3.35 Å apart.

reading is that a bilayer’s lower layer can dish downward and absorb part of the gas work, so less of it goes into opening new interface — but that is an interpretation of two pressures, not a measured threshold, and it deserves a proper sweep before being believed.

6 How to validate a toy model against LAMMPS

6.1 The tests, strongest first

1. **Same state, same energy.** Export the exact configuration and compare total potential energy. This is by far the strongest test: it is a single number summing every interaction, and agreement to six figures is very hard to obtain by accident. Everything else is weaker.
2. **Known physical constants.** Cohesive energy per atom, lattice constant, elastic moduli, phonon frequencies. These catch a mis-parameterised potential that is nonetheless self-consistent.
3. **An analytic limit.** Hencky, a harmonic oscillator, an isolated dimer — anything with a closed-form answer.

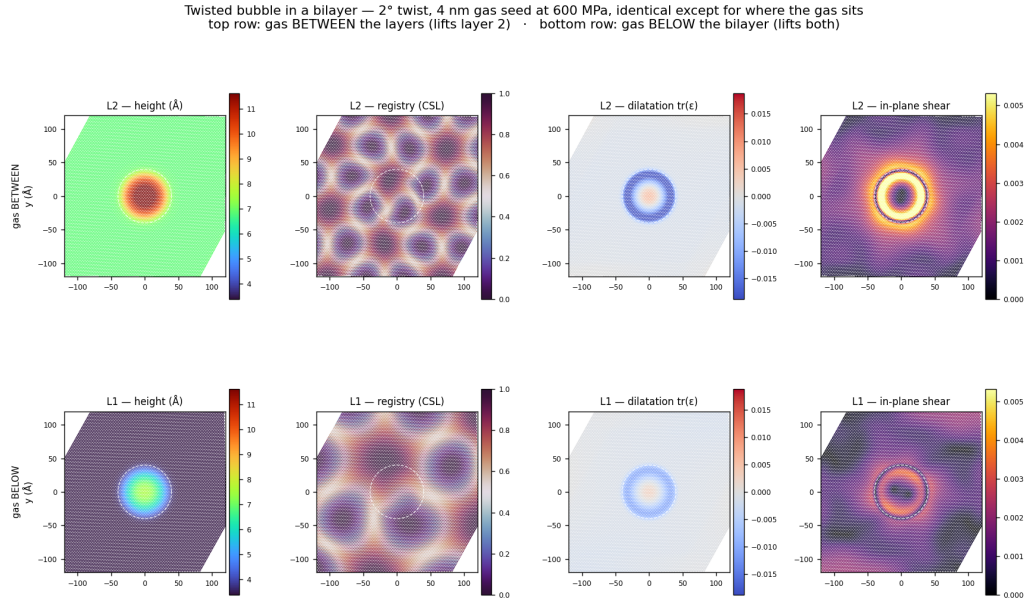


Figure 3: Height, registry, dilatation and shear for the driven layer in each placement. The strain rings sit at the seed radius in every panel.

4. **Observable geometry.** Heights, profiles, aspect ratios. Weaker, because many wrong models produce plausible shapes.
5. **Trajectory agreement.** Effectively worthless. MD is chaotic; correct implementations diverge atom-for-atom within hundreds of steps, and float32 versus float64 is enough to do it.

6.2 Five ways a validation lies

Each of these occurred while preparing this note.

The vacuum test. A check that “no surviving bond exceeds the break threshold” passed with a maximum strain of 0.0000 — because every bond had broken and the set was empty. *Always assert that the set under test is non-empty*, and prefer conditions that would fail if the system were destroyed.

The unconverged measurement. A pressure scan reported an aspect ratio of 0.083 at $R = 10$ nm, suggesting the size-independence of (6) was violated. The runs were time-limited to ~ 4700 steps. Given ten times as long the same case reached 0.129 and was still rising toward the predicted 0.134. *A quantity that has not stopped changing is not a measurement.*

The shared term. Toy model and AIREBO agreed on blister height to 0.5 %, which looks like strong mutual validation. But the two engines differ *only* in the in-plane potential: they share the same Lennard-Jones substrate adhesion and the same gas model, both of which live in the plugin rather than in the potential. So the agreement validates the membrane half and says nothing about adhesion — and adhesion is precisely what sets the boundary condition responsible for the 31 % Hencky discrepancy. *Two models agree most easily about the parts they share.*

The statistic that cannot see the answer. A settling test built on the difference between two window means reported 13% “drift” for a system whose mean was flat to 0.55 Å over 35 000 steps. It was measuring a decaying oscillation, which that statistic cannot distinguish from a trend. *Choose a convergence statistic that matches how the system actually approaches equilibrium* — for a ringing system, fit the trend and track the amplitude separately.

The proxy that stops meaning what it meant. A stability criterion written as “the blister centre is below the Lennard-Jones cutoff” was sound for a 3 nm blister and nonsense for a 10 nm one, where a correct blister is necessarily taller than the cutoff. It silently relabelled every large blister as “escaped”. *A threshold derived for one regime must be re-derived, not rescaled, for another.*

6.3 A working procedure

1. Build the configuration once, in the fast model.
2. Export it and confirm the energy matches to five or six figures. If it does not, stop — nothing downstream is meaningful.
3. Identify which terms the two codes *share*, and state explicitly that those are untested by the comparison.
4. Compare a physical observable, with a convergence criterion measured, not assumed.
5. Check against an analytic limit where one exists, and when it disagrees, determine whether the model or the analytic assumption is at fault.
6. Use the fast model for exploration and the expensive one for spot checks at the settings that matter.

7 Practical notes

Tooling for the above lives in `DSW/tools/`: `cyclerun.js` (build, drive, export, capture), `gassweep.js` and `gasconfirm.js` (blister equilibria and Hencky comparison), `registry.py` (registry computed identically for plugin and LAMMPS data), `makemovies.py` and `blister_movie.py`. `DSW/tools/README.md` lists the traps that cost real time.

For simulating a membrane in a material with no classical potential — α -RuCl₃ being the case at hand — the ordering follows from Section 1.3. Blister mechanics needs only E_{2D} , bending rigidity and Γ , so refitting the toy model to those three numbers gives a usable membrane quickly and will be honestly wrong about chemistry. A Stillinger–Weber fit to DFT phonons is the middle route, and the three-type reduced realisation already used for WS₂, MoTe₂ and WTe₂ applies unchanged to a Cl–Ru–Cl trilayer. A universal machine-learned potential is the best fidelity available. None of them is magnetic, which for a Kitaev material is a real limitation.

A A worked LAMMPS deck, line by line

This is not an illustrative script. It is the deck the plugin exported for a **twisted bubble** — 2° twist, 7.5 nm Hencky gas pocket at 100 kPa, 232 497 atoms — and it is the one that was actually run: 7 min 06 s on 18 threads, 226 dump frames, no errors. Its first line of output is the check that matters:

plugin E_{pot} at the moment of export	−709090 eV
LAMMPS PotEng at step 0 of this deck	−709089.85 eV

Six significant figures, from two independent codes evaluating the same configuration. If any line below were wrong, that number would not match.

Listing 1: run.in as exported, verbatim.

```

1 # graphene-md -> LAMMPS export (embedded-LAMMPS engine, graphene), continue-from-now
2 # plugin state at export: frame 5, Epot -709090 eV, T 0.1 K
3 # Run from this directory: lmp -in run.in
4 #
5 # THREADS: LAMMPS defaults to ONE OpenMP thread when OMP_NUM_THREADS is
6 # unset, so an idle multicore box will quietly run serial. Uncomment the
7 # two lines below (set N to your physical core count) or export
8 # OMP_NUM_THREADS before launching. Needs a build with PKG_OPENMP.
9 package omp 18
10 suffix omp
11
12 units metal
13 dimension 3
14 boundary m m m # 'm', never 's': shrink-wrap on a flat sheet kills neighbor binning
15 atom_style atomic
16 read_data system.data
17
18 # The plugin computed the sheet with airebo and added the substrate LJ
19 # through fix external; here that becomes a native hybrid cross term.
20 pair_style hybrid airebo 3.0 lj/cut 10.26
21 pair_coeff * * airebo "C:/Users/pbog/Dropbox/ACTIVITIES/00 VSCODE/DSW/Plugins/2D Materials/graphene-md/
    potentials/CH.airebo" C NULL
22 pair_coeff 1 2 lj/cut 0.002387 3.42 10.26
23 pair_coeff 2 2 none
24
25 neighbor 2.0 bin
26 neigh_modify every 1 delay 0 check yes
27 timestep 0.001
28
29 group sheet type 1
30 group sub type 2
31 group edge empty # edge springs are off (edgeK = 0); membership not computed
32
33 # substrate is rigid: simply not integrated
34 fix mdint sheet nve
35 fix visc sheet viscous 1.244852422e-05
36
37 # ----- protrusion driver -----
38 variable tps equal elapsed*dt
39 # ramp-hold-return continuing from elev=0.05 A: up to 6.922 A at 10 A/ps,
40 # hold 0.5 ps, return; piecewise via boolean algebra
41 variable elev equal "(0.05+10*v_tps)*(v_tps<0.68720534)+6.9220534*(v_tps>=0.68720534)*(v_tps<1.1872053)
    +(6.9220534-10*(v_tps-1.1872053))*(v_tps>=1.1872053)*(v_tps<1.8794107)"
42 variable gasF equal "0.040881825*((v_elev/6.9220534)*((v_elev/6.9220534)<1)+((v_elev/6.9220534)>=1))"
43 variable wfoot atom "(((x-0)^2+(y-0)^2)<5625)*((1-(((x-0)^2+(y-0)^2)/5625)*(((x-0)^2+(y-0)^2)<5625)))
    ^0.6666667)"
44 variable fzg atom v_gasF*v_wfoot
45 fix gas sheet addforce 0.0 0.0 v_fzg
46
47 # Temperature of the SHEET, not the box: thermo's default temp averages
48 # over the frozen substrate too and reads roughly half the real value.
49 compute Tsheet sheet temp
50 thermo 100
51 thermo_style custom step time pe ke c_Tsheet press
52 # Large system: the sheet is dumped often, the (rigid) substrate rarely.
53 # Dumping every atom this often would be gigabytes of text.
54 dump sh sheet custom 10 sheet.lampstrj id x y z
55 dump_modify sh sort id
56 dump sb sub custom 250 sub.lampstrj id x y z
57 dump_modify sb sort id
58
59 # The ramp-hold-return completes at step 1879; the tail lets the
60 # sheet settle once the protrusion is fully retracted.
61 run 2254

```

What each part does

Lines 5–11, threads. LAMMPS silently falls back to *one* OpenMP thread when `OMP_NUM_THREADS` is unset, which quietly wastes an 18-core machine. `suffix omp` then selects the `/omp` variants of every style that has one.

Line 13, units metal. Energies in eV, distances in Å, time in ps, so `timestep 0.001` is 1 fs. Every number below is in those units.

Line 15, boundary m m m. Non-periodic with a *minimum* box rather than shrink-wrapped (`s`). This matters: on a flat sheet, shrink-wrapping collapses the box in z to near-zero thickness and LAMMPS refuses to bin the neighbour list.

Line 16, atom_style atomic. No bond list at all. AIREBO determines bonding from geometry every step, so there is nothing to declare — the difference from the classic engine’s `molecular` style, which does carry an explicit bond and angle list.

Lines 20–23, the potential.

- `airebo 3.0` — the 3.0 is the LJ cutoff in units of σ ; the adaptive LJ and torsion terms are on by default.
- `pair_coeff * * airebo CH.airebo C NULL` — one element per atom type. Type 1 is carbon; type 2 is NULL, i.e. the substrate takes no part in AIREBO at all.
- `pair_coeff 1 2 lj/cut 0.002387 3.42 10.26` — the sheet’s adhesion to the substrate, $\varepsilon = 2.387$ meV, $\sigma = 3.42$ Å, cut at 3σ . This replaces the `fix external` callback the plugin uses internally.
- `pair_coeff 2 2 none` — the substrate does not interact with itself. It is rigid, so that energy is a constant and computing it would only cost time.

Lines 29–31, groups. `sheet` and `sub` by type. The `edge` group is empty here because the edge-spring collar is switched off; it is emitted as `group edge empty` rather than as a list of atom ids, because `group ...id` rescans every atom once per argument and a 37 000-id list once kept LAMMPS in setup for over twelve minutes.

Lines 33–35, integration. **Only the sheet is integrated** — there is no `fix nve` on `sub`, which is exactly how a rigid substrate is expressed. `fix viscous` supplies the plugin’s damping. Note this is *not* a thermostat: it removes energy without supplying fluctuations, so the run relaxes toward $T = 0$ rather than sampling an ensemble.

Lines 37–43, the protrusion driver. The interesting part, and the one worth checking arithmetic on.

- `v_tps` is elapsed time in ps.
- `v_elev` is a ramp–hold–return written as boolean algebra, because LAMMPS equal-style variables have no `if`. The three terms are $(\text{rise}) \times (t < t_1) + (\text{hold}) \times (t_1 \leq t < t_2) + (\text{fall}) \times (t_2 \leq t < t_3)$. With a rise rate of 10 Å/ps (the plugin’s 0.01 Å/step over a 1 fs step) from 0.05 Å to the Hencky height 6.922 Å: $t_1 = 0.6872$ ps, a 0.5 ps hold gives $t_2 = 1.1872$, and the fall gives $t_3 = 1.8794$ ps. That is 1879 steps, which is exactly the cycle length quoted at line 60.

- `v_gasF` is the force per atom at full pressure, $0.0408818 \text{ eV}/\text{\AA}$. Check it: $10^8 \text{ Pa} \times 6.2415 \times 10^{-12} \times 2.62 \text{ \AA}^2 \times 25 = 0.04088$. **The 25 is betweenBoost**, a deliberate fudge in the open-loop model without which a 1 bar pocket does nothing at this scale. It is not physics, and the fixed- N model in Section 3.4 does not use it.
- `v_wfoot` is the Hencky footprint $(1 - r^2/R^2)^{2/3}$ with $R^2 = 5625 \text{ \AA}^2$, i.e. $R = 75 \text{ \AA}$. The nested $((r^2 < R^2))$ inside the base is not redundant: outside the blister $1 - r^2/R^2$ is negative, a negative base to a fractional power is NaN, and $0 \times \text{NaN} = \text{NaN}$. Folding the guard into the base keeps it at exactly 1 outside. An earlier version guarded only the result and NaN'd the entire sheet on the first step.
- `fix gas sheet addforce` applies it, upward only.

Lines 45–48, thermodynamics. `compute Tsheet sheet temp` is not cosmetic: thermo's default temperature averages over the frozen substrate as well, and with roughly 60 % of the atoms frozen it reads about half the true sheet temperature.

Lines 49–54, output. The sheet is dumped ten times more often than the substrate. Dumping every atom at the sheet's cadence would be gigabytes of text for no extra information, since the substrate is rigid.

What this deck does *not* reproduce

Worth stating plainly, because it is what a reader would otherwise assume.

1. **This is the open-loop gas model.** Pressure follows the schedule regardless of the bubble's volume, so it inflates without the $p = NkT/V$ feedback of Section 3.4, and the `betweenBoost` factor of 25 is active. It faithfully reproduces what the plugin did in this run — it is not the better model.
2. **The footprint is a fixed disc.** The free peel front is *not* exported: the exporter emits the fixed-radius form, so a deck cannot currently reproduce a moving contact line. Since Γ is defined at a moving front, no adhesion energy should be extracted from this deck.
3. **No bilayer.** This is one mobile sheet on a rigid substrate. The two-live-layer runs come from a different plugin (`moire-bubble`) which has no exporter at all, so those results exist only inside DSW.
4. **Velocities are carried but the state is not a thermal ensemble.** `system.data` includes a `Velocities` section so the run continues from the plugin's instant, but with `fix viscous` and no thermostat the trajectory relaxes rather than samples.
5. **Classic-engine decks omit the bending umbrella.** Not an issue here — this is the AIREBO engine, where bending is carried by the potential — but a deck exported from the Morse engine loses that term, which is why its blisters sit lower.